

INTERNATIONAL STANDARD

Radiation protection instrumentation – Alarming personal radiation devices (PRDs) for the detection of illicit trafficking of radioactive material





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INTERNATIONAL STANDARD

Radiation protection instrumentation – Alarming personal radiation devices (PRDs) for the detection of illicit trafficking of radioactive material

INTERNATIONAL
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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**RADIATION PROTECTION INSTRUMENTATION – ALARMING PERSONAL
RADIATION DEVICES (PRDs) FOR THE DETECTION OF ILLICIT
TRAFFICKING OF RADIOACTIVE MATERIAL**

FOREWORD

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International Standard IEC 62401 has been prepared by subcommittee 45B: Radiation protection instrumentation, of IEC technical committee 45: Nuclear instrumentation.

This second edition cancels and replaces the first edition of IEC 62401, issued in 2007. It constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) making the standard consistent with the new standards for detection of illicit trafficking of radioactive material (see the Introduction);
- b) changing some requirements:
 - removal of the 2 levels of background levels (high and low) needed for the different tests. Only one background level (laboratory) remains,
 - the gamma alarm is tested using moving sources and not statically (6.2),

- relative intrinsic error,
 - over-range,
 - detection of neutrons;
- c) creating a uniform functionality test for all environmental, electromagnetic and mechanical tests and a requirement for the coefficient of variation of each nominal mean reading;
- d) reference to IEC 62706 for the environmental, electromagnetic and mechanical test conditions.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
45B/881/FDIS	45B/888/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

Illicit and inadvertent movement of radioactive materials has become a problem of increasing importance. Radioactive sources out of regulatory control, so-called “orphan sources”, have frequently caused serious radiation exposures and widespread contamination. Although illicit trafficking in nuclear and other radioactive materials is not a new phenomenon, concern about a nuclear “black market” has increased in the last few years, particularly in view of its terrorist potential.

In response to the technical policy of the International Atomic Energy Agency (IAEA), the World Customs Organization (WCO) and the International Criminal Police Organization (Interpol) related to the detection and identification of special nuclear materials and security trends, nuclear instrumentation companies are developing and manufacturing radiation instrumentation to assist in the detection of illicit movement of radioactive and special nuclear materials. This type of instrumentation is widely used for security purposes at nuclear facilities, border control checkpoints, and international seaports and airports.

However, to ensure that measurement results made at different locations are consistent, it is imperative that radiation instrumentation be designed to rigorous specifications based upon agreed performance requirements stated in international standards. Several IEC standards have been developed to address body-worn, hand-held and portal instruments, see Table 1.

Table 1 – IEC standards concerning instruments for the detection of illicit trafficking of radioactive material

Type of instrumentation	IEC number	Title of the standard
Body-worn	62401	Radiation protection instrumentation – Alarming Personal Radiation Devices (PRDs) for the detection of illicit trafficking of radioactive material
	62618	Radiation protection instrumentation – Spectroscopy-Based Alarming Personal Radiation Devices (SPRD) for detection of illicit trafficking of radioactive material
	62694	Radiation protection instrumentation – Backpack-type radiation detector (BRD) for detection of illicit trafficking of radioactive material
Portable or hand-held	62327	Radiation protection instrumentation – Hand-held instruments for the detection and identification of radionuclides and for the estimation of ambient dose equivalent rate from photon radiation
	62533	Radiation protection instrumentation – Highly sensitive hand-held instruments for photon detection of radioactive material
	62534	Radiation protection instrumentation – Highly sensitive hand-held instruments for neutron detection of radioactive material
Portal	62244	Radiation protection instrumentation – Installed radiation portal monitors (RPMs) for the detection of illicit trafficking of radioactive and nuclear materials
	62484	Radiation protection instrumentation – Spectroscopy-based portal monitors used for the detection and identification of illicit trafficking of radioactive material
Data format	62755	Radiation protection instrumentation – Data format for radiation instruments used in the detection of illicit trafficking of radioactive materials

RADIATION PROTECTION INSTRUMENTATION – ALARMING PERSONAL RADIATION DEVICES (PRDs) FOR THE DETECTION OF ILLICIT TRAFFICKING OF RADIOACTIVE MATERIAL

1 Scope

This document applies to alarming radiation detection instruments that are pocket-sized, carried on the body and used to detect and indicate the presence and general magnitude of gamma radiation fields. Neutron detection may also be provided.

Personal Radiation Devices (PRDs) alert the user to the presence of a source of radiation that is distinctly above the measured average local background radiation level. They are not intended to provide a measurement of the ambient or personal dose equivalent rate.

The object of this document is to describe design and functional criteria along with testing methods for evaluating the performance of the PRDs used for detection of illicit trafficking of radioactive material (e. g., for border radiation monitoring).

This document does not apply to the ambient or personal dose equivalent rate meters which are covered in IEC 60846-1 or IEC 61526, respectively. If the manufacturer states that the PRD can be used for radiation protection purposes, compliance with IEC 60846-1 or IEC 61526 will be needed.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-395, *International Electrotechnical Vocabulary – Part 395: Nuclear instrumentation: Physical phenomena, basic concepts, instruments, systems, equipment and detectors*

IEC 60068-2-11, *Basic environmental testing procedures – Part 2-11: Tests – Test Ka: Salt mist*

IEC 62706:2012, *Radiation protection instrumentation – Environmental, electromagnetic and mechanical performance requirements*

IEC 62755, *Radiation protection instrumentation – Data format for radiation instruments used in the detection of illicit trafficking of radioactive materials*

3 Terms and definitions, abbreviated terms and symbols, quantities and units

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-395, as well as the following, apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1.1

A-weighted sound level

frequency weighting of an acoustic spectrum according to a standardized frequency response curve based on the frequency response of the human ear

3.1.2

accuracy

quality which characterizes the ability of a measuring instrument to provide an indicated value close to a true value of the measurand

3.1.3

alarm

audible, visual, or other signal activated when the instrument reading exceeds a pre-set value or falls outside of a pre-set range

3.1.4

background

radiation field in which there are no external sources present other than those in the natural radiation field at the location of the measurements

3.1.5

coefficient of variation

COV

ratio of the standard deviation s to the arithmetic mean $\bar{x}$ of a set of n measurements x_i given by the following formula:

$$COV = \frac{s}{\bar{x}} = \frac{1}{\bar{x}} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

3.1.6

conventionally true value of a quantity

best estimate of the value of a quantity used for a given purpose

3.1.7

effective range of measurement

range of values of the quantity to be measured over which the performance of a device meets the requirements of this document

3.1.8

false alarm

alarm not caused by a radioactive source under the specified background conditions

3.1.9

fluence

quotient of dN by da , where dN is the number of particles incident on a sphere of cross-sectional area da (unit: m^{-2})

3.1.10

fluence rate

quotient of $d\Phi$ by dt , where $d\Phi$ is the increment of the fluence in the time interval dt (unit: $\text{m}^{-2} \cdot \text{s}^{-1}$)

3.1.11**point of measurement**

place at which the conventionally true value is determined

3.1.12**reference point**

defined position on instrument to be used to position it at a point where the conventionally true value of the quantity to be measured is known

3.1.13**relative intrinsic error** ε_{REL}

difference between the instrument's reading, M , and the conventionally true value, CTV , of the quantity being measured divided by the conventionally true value when subjected to a specified reference quantity under specified reference conditions

$$\varepsilon_{\text{REL}} = \frac{M - CTV}{CTV}$$

3.1.14**response**

reading or indication of the PRD

3.1.15**standard deviation**

positive square root of the variance

3.1.16**standard test conditions**

prescribed range for influence quantities to be used during testing of a measuring instrument

3.1.17**type test**

test for conformity evaluation on the basis of one or more specimens of a product representative of the production

3.1.18**routine test**

test to which an individual device is subjected during or after manufacture to ascertain whether it complies with certain criteria

3.1.19**acceptance test**

contractual test to prove to the customer that the device meets certain conditions of its specification

3.1.20**uncertainty <of measurement>**

parameter, associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand

Note 1 to entry: Uncertainty of measurement comprises, in general, many components. Some of these components may be evaluated from statistical distribution of the results of series of measurements and can be characterized by experimental standard deviations. The other components, which can also be characterized by experimental standard deviations, are evaluated from assumed probability distributions based on experience or other information.

Note 2 to entry: It is understood that the result of the measurement is the best estimate of the value of the measurand and that all components of uncertainty, including those arising from systematic effects, such as components associated with corrections and reference standards, contribute to the dispersion.

3.1.21

variance

σ^2

measure of dispersion, which is the sum of the squared deviation of observations x_i from their mean $\bar{x}$ divided by one less than the number of observations n

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

3.2 Abbreviated terms and symbols

COV	coefficient of variation
EMC	Electromagnetic Compatibility
ESD	electrostatic discharge
IP	degree of ingress protection
LED	light-emitting diode
PRD	personal radiation device
RF	radio frequency

3.3 Quantities and units

In the present document, units of the International System (SI) are used¹. The definitions of radiation quantities are given in IEC 60050-395.

Nevertheless, the following units may also be used:

- for energy: electron-volt (symbol: eV), 1 eV = 1,602 × 10⁻¹⁹ J;
- for time: hours (symbol: h) and minutes (symbol: min).

Multiples and submultiples of SI units will be used, when practicable, according to the SI system.

4 General characteristics and requirements

4.1 General

PRDs tested using this document are carried on the body and are used to detect and indicate the presence and general magnitude of the gamma radiation fields (e.g., $\mu\text{Sv}\cdot\text{h}^{-1}$, counts per second, numerical display without units (1-9), LEDs). PRDs are not intended to provide a measurement of the ambient or personal dose equivalent rate. However, the manufacturer may provide an optional display of the ambient or personal dose equivalent rate from gamma radiation.

The following are important design features:

- quickly alert the user to small increases of radiation levels with a low occurrence of false alarms;
- simple to use for personnel not expert in radiation measurements;

¹ International Bureau of Weights and Measures: The international System of Units (SI), 8th edition, 2014.

- separate gamma and neutron radiation alarms (if neutron response is provided), with visual and audible alerts;
- audible and/or visual indication that corresponds to the magnitude of the radiation field (e.g., increasing frequency or pitch of beep tone with increasing radiation signal);
- the visual display is readable in low light levels (< 150 lx) and high light levels (>10 000 lx);
- small, rugged, shock-resistant, water-resistant and dust resistant;
- protection of the setting of all operational parameters;
- self-diagnostic capabilities (e.g., detector fault);
- indication of battery status;
- vibration alarm and/or earphone with user-adjustable volume;
- personal protection alarm with pre-settable level;
- the operating parameters and the recorded data are not lost if there is a loss of power.

4.2 Mechanical requirements

4.2.1 Size

The overall dimensions of the PRD, excluding any clip, retaining device or external alarm, should not exceed 15 cm in length, 10 cm in width and 5 cm in thickness.

4.2.2 Mass

The mass of the complete instrument including battery, clip, etc., should not exceed 400 g.

4.2.3 Alarm characteristics

- a) Indicator – The PRD shall alert the user when an alarm occurs (e.g., visual indicator that enables the operator to view without removal of the device from its wearing position, vibration alarm intensity that can be sensed by the user, etc.).
- b) Alarm type – The PRD shall have audible, visual and vibratory alarms. The frequency of an audible alarm should be within the range of 1 kHz to 4 kHz. Where an intermittent alarm is provided, the signal interval shall not exceed 2 s. The alarm volume at a distance of 30 cm from the alarm source shall be at least 85 dB(A). The A-weighted sound level shall not exceed 100 dB(A) at 30 cm from the alarm source. If the PRD provides neutron detection, the neutron alarm shall be different from the gamma alarm.
- c) The personal protection alarm shall be distinguishable from the other alarms.
- d) The PRD audible and vibratory alarms shall be capable of being acknowledged and silenced by the PRD user.

4.2.4 Case construction

The PRD case should be smooth, rigid, resistant to mechanical shock, dust-resistant and water-resistant. Means shall be provided to securely affix the instrument to the user (for example, a clip, holster or ring), with attention given to the necessary orientation of the detector, alarm type and display. See 7.3 for the IP classification.

4.2.5 Reference point marking

The reference point of the PRD shall be marked or described in the manual.

4.2.6 Switches

External switches shall be adequately protected to prevent accidental or unauthorized operation.

4.3 Data format

If the PRD is capable of transferring data, the data format should meet the IEC 62755 requirements. As a minimum the data shall contain the following information:

- manufacturer name;
- instrument model;
- serial number;
- software version;
- gamma detector kind (e.g., CsI(Tl), NaI(Tl));
- date and time of measurement;
- measured radiation levels (e.g., count rate, dose rate, unit-less level);
- gamma-ray alarm indication;
- personal protection alarm.

If the PRD is equipped with a neutron detector, in addition the data shall contain the following information:

- neutron detector kind (e.g., ^3He , Lithium Glass, BF_3);
- neutron level (e.g., count rate);
- neutron alarm indication.

The data transfer protocol shall be fully described by the manufacturer. Consideration should be given to data security when using wireless data transfer techniques.

4.4 Explosive atmospheres

The manufacturer shall state whether or not the PRD is certified for use in explosive atmospheres and its category. Proof of certification shall be provided when claimed. Certification should be based on IEC 60079-11 or equivalent standard (e.g. UL-913).

5 General test procedures

5.1 Nature of tests

The required standard test conditions for environmental quantities, such as temperature and humidity, as well as those for other quantities that may influence the performance of the PRDs, are given in Table 2. Acceptable testing ranges for these quantities shall be met, except where the effect of the condition or quantity itself is being tested.

Table 2 – Standard test conditions

Influence quantities	Standard test conditions (unless otherwise indicated by the manufacturer)
Ambient temperature	18 °C to 25 °C ^a
Relative humidity	≤ 75 % ^a
Atmospheric pressure	96 kPa to 106 kPa ^a
Gamma radiation background	Less than ambient dose equivalent rate of 0,15 $\mu\text{Sv}\cdot\text{h}^{-1}$
Neutron radiation background	Neutron fluence rate less than 200 $\text{s}^{-1}\cdot\text{m}^{-2}$
^a The values are intended for tests performed in temperate climates. In other climates, the actual values of the quantities at the time of test shall be stated. Similarly, a lower limit of pressure of 70 kPa may be permitted at higher altitudes.	

The tests in this document are to be considered as type tests, unless otherwise stated. The user may employ parts of the document as acceptance tests.

All tests using this document shall be performed in the configuration of intended use. Functional settings used by the PRD shall not be changed during the different tests if this is not specified in the corresponding method of test.

Where no method of test is described for a requirement, it is understood to mean that either the information for the method of test is contained in the requirement or the characteristic requirement can be verified by observation or by consultation of the manufacturer's specifications.

5.2 Statistical fluctuations

When performing environmental, electromagnetic or mechanical tests, the coefficient of variation (COV) of the readings shall be less than or equal to 12 %. If the COV is larger than 12 %, then the radiation field or the number of readings should be increased to ensure that the mean value of such readings may be estimated with sufficient accuracy to demonstrate compliance with the test in question.

For neutron or gamma background measurements, attaining a COV to meet this requirement may not be possible. Therefore, testing with neutrons or gammas at background levels (i.e., testing without radioactive source present) may be performed even when the COV is larger than 12 %.

The requirement concerning the COV does not apply for the PRDs with unit-less display.

5.3 General test information

The reference point of the PRD shall be placed at the point of measurement. The instrument shall be oriented with respect to the direction of the radiation field as indicated by the manufacturer.

The following radionuclides shall be used for gamma testing: ^{241}Am , ^{137}Cs and ^{60}Co . For neutrons, the test source shall be ^{252}Cf moderated by being surrounded by a spherical shell with 4 cm wall thickness of high density polyethylene or equivalent moderator and inner cavity diameter of not more than 3 cm.

When performing the radiation tests in Clause 6, the PRD shall be mounted on the centre of a phantom to simulate the human torso. The phantom shall be made of polymethyl methacrylate (PMMA). The phantom dimensions shall be 30 cm wide, 30 cm high and 15 cm thick. The reference point of the PRD shall be placed at a height of at least 1 m from the floor.

5.4 Instrument setup

The PRD to be tested shall be placed under standard test conditions, switched on, set up following instructions from the manufacturer, and allowed a stabilisation and background detection period specified by the manufacturer's recommendations. The entire process from the time the PRD is turned on until it is ready for normal operation should not exceed 1 min.

5.5 Speed of moving sources

For dynamic tests, the source or PRD shall be moved in a configuration that provides no shielding around the source other than that required for the specific test. The source speed shall be $1,2 \text{ m}\cdot\text{s}^{-1} \pm 10 \%$ (average walking speed). The nominal distance of closest approach $d_0 = 1,5 \text{ m}$, unless otherwise required in a test. If the distance of closest approach, d (expressed in m), is adjusted in order to maintain the same ambient dose equivalent rate at the reference point of the PRD then the passage speed, v (expressed in $\text{m}\cdot\text{s}^{-1}$), shall be adjusted to:

$$v = v_0 \times d/d_0$$

where

$v_0 = 1,2 \text{ m} \cdot \text{s}^{-1}$, and

$d_0 = 1,5 \text{ m}$.

It is recommended that the testing distance of closest approach is kept between 1 m and 3 m.

During the dynamic tests, there shall be a 10 s minimum delay between each trial with the source either positioned at a distance where it does not affect the background surrounding the PRD or shielded during the delay.

5.6 Functionality test

5.6.1 General

The functionality test consists of comparing the PRD functionality before (pre-test) and during or after (post-test) an environmental, mechanical or electromagnetic disturbance described in Clauses 7, 8 and 9.

5.6.2 Pre-test (preparation)

The pre-test or preparation includes the stabilisation of the PRD in normal test conditions and verification of its correct functioning (alarming at an increase of at least $0,5 \mu\text{Sv} \cdot \text{h}^{-1}$ over the ambient background level of radiation). Then, the radiation field should be set using a gamma source in order to obtain a COV of the readings less than or equal to 12 % (see 5.2). Record at least 10 gamma indications (if provided). Calculate the mean value of the PRD gamma readings.

The PRD indication can be a unit indication (e.g., $\mu\text{Sv} \cdot \text{h}^{-1}$, counts/s, etc.) or a unit-less indication (e.g., display 1-9, LEDs, etc.).

If neutron detection is provided, place a neutron source at a position where the PRD alarms. For this, place first the neutron source at a position where the PRD does not alarm and bring closer progressively the neutron source until you find a position where the PRD alarms.

5.6.3 Post-test

The test has two parts:

a) alarm check

In the appropriate part of Clauses 7, 8 or 9, it is stated whether the functionality test is performed during or after the environmental, mechanical or electromagnetic disturbance. Thus, during or after a disturbance (e.g., the low and high temperature extremes), the PRD shall be exposed to a gamma source producing an ambient dose equivalent rate of at least $0,5 \mu\text{Sv} \cdot \text{h}^{-1}$ over the ambient background level of radiation. The instrument shall alarm. If the neutron indication is provided, place the neutron source at the same position determined according to 5.6.2 and check that the neutron alarm activates.

There shall be no alarm during the test when radioactive sources are not present, unless otherwise specified.

b) indication check

Place the gamma source at the same position as in the pre-test and take the same number of readings as in the pre-test. The mean value of the post-test readings shall be calculated and compared to the mean value of the pre-test readings. The mean value shall not change by more than:

- ± 30 % (for the unit indications),
- or one neighbouring indication change (for the unit-less indication).

NOTE An example of an accepted neighbour change of the unit-less indication is the display change from 5 to 6 (or 4) or turning on/off of one LED.

There shall be no visible external damage after the environmental, mechanical or electromagnetic disturbance and the instrument shall function correctly as prior to these tests.

6 Radiation detection requirements

6.1 Rate of false alarms

6.1.1 Requirements

The number of false alarms shall be not greater than 1 alarm in 1 h. If the PRD has neutron detection capabilities the same maximum number of neutron false alarms applies.

6.1.2 Method of test

The PRD shall be monitored at the background radiation level for 8 h. The number of gamma false alarms during this period shall not exceed 3.

NOTE This corresponds to 95 % one-sided upper confidence bound for 1 false alarm for 1 h in Poisson distribution.

If the PRD has neutron detection capabilities, the same maximum number of neutron false alarms applies.

6.2 Gamma alarm

6.2.1 Requirements

The PRD shall alarm within 2 s after exposure to an increase in the ambient dose equivalent rate level of $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$ (-0 %, $+20$ %) above background for ^{241}Am , ^{137}Cs and ^{60}Co sources when moving at a speed of $1,2 \text{ m}\cdot\text{s}^{-1}$ (± 10 %) at a distance of closest approach of 1,5 m.

6.2.2 Method of test

The PRD shall be switched on and allowed to stabilise as described in the manufacturer's manual without any radiation sources present in the testing area.

Move the ^{137}Cs source towards and past the PRD as described in 5.5. The speed of $1,2 \text{ m}\cdot\text{s}^{-1}$ shall be scaled using the information provided in this subclause, if the distance of closest approach is different from 1,5 m.

The ambient dose equivalent rate at the reference point of the PRD shall be increased by $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$ (-0 %, $+20$ %) above background when the ^{137}Cs source is at the distance of closest approach.

The gamma alarms shall activate not later than 2 s after the source passes through the point that is closest to the PRD. The response is also acceptable if the gamma alarm is activated before the maximum radiation field is reached (i.e., before reaching the required test position). The radiation field shall be reduced (see 5.5) and the test repeated 19 additional times for a total of 20 trials.

The entire process shall be repeated using ^{241}Am and ^{60}Co sources, each producing an ambient dose equivalent rate of $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$ ($+20$ %) above background at the reference point of the PRD at the distance of closest approach.

The test results are acceptable if the PRD alarms in at least 19 out of 20 trials for each source.

6.3 Personal protection alarm

6.3.1 Requirements

The PRD shall alarm within 2 s of exposure to an increase in the ambient radiation level greater than personal protection alarm threshold.

6.3.2 Method of test

For this test, set the personal protection alarm threshold to at least $10 \mu\text{Sv}\cdot\text{h}^{-1}$. Using a ^{137}Cs source, increase the ambient dose equivalent rate at the reference position to 30 % above the personal protection threshold. This shall trigger a personal protection alarm in less than or equal to 2 s.

6.4 Relative intrinsic error

6.4.1 Requirements

This only applies where the PRD has an ambient dose equivalent rate display.

Under standard test conditions, the relative intrinsic error in the response of the instrument to the reference photon radiation from ^{241}Am , ^{137}Cs and ^{60}Co shall not exceed ± 50 % for all ambient dose equivalent rates over the manufacturer's stated range.

6.4.2 Method of test

The test shall be performed using ^{241}Am , ^{137}Cs and ^{60}Co sources for at least 3 ambient dose equivalent rates: one at approximately $1 \mu\text{Sv}\cdot\text{h}^{-1}$, one at approximately $10 \mu\text{Sv}\cdot\text{h}^{-1}$ and one at approximately 50 % of the maximum stated range of indication. The relative intrinsic error of the instrument's mean response shall not exceed ± 50 % for each of the 3 radionuclides.

6.5 Detection of gradually increasing radiation levels

6.5.1 Requirements

The PRD's alarm threshold shall not be affected by slowly increasing radiation levels that may be caused when a wearer is slowly approaching or is being approached by a radiation source.

6.5.2 Method of test

The PRD shall be switched on and allowed to stabilise as described in the manufacturer's manual without any radiation sources present in the testing area.

From a distance of 5 m away from the PRD, move the ^{137}Cs source towards the PRD as described in 5.5. The speed of $0,1 \text{ m}\cdot\text{s}^{-1} \pm 10$ % shall be scaled using the information provided in 5.5, if the distance of closest approach is different from 1,5 m.

The ambient dose equivalent rate at the reference point of the PRD shall be increased by $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$ (-0 %, $+20$ %) above background when the ^{137}Cs source is at the distance of closest approach.

The gamma alarms shall activate not later than 2 s after the source passes through the point that is closest to the PRD. The response is also acceptable if the gamma alarm is activated before the maximum radiation field is reached (i.e., before reaching the required test position). The field shall be reduced (see 5.5) and the test repeated 2 additional times for a total of 3 trials. Wait at least 100 s between the different trials.

The test results are acceptable if the PRD alarms in 3 out of 3 trials.

6.6 Over-range

6.6.1 Requirements

The manufacturer shall state the ambient dose equivalent rate at which the PRD indicates the over-range condition. The PRD shall indicate that an over-range condition exists when the ambient dose equivalent rate is greater than the manufacturer-stated over-range. During the over-range condition, the alarm indication shall stay activated. The PRD shall recover within 1 min after removing the over-range exposure.

6.6.2 Method of test

Using a gamma source, apply a step change in the ambient dose equivalent rate, from background to 2 times the value that triggers the over-range condition, as stated by the manufacturer, and expose the PRD for 5 min.

The PRD shall indicate an over-range condition within 5 s after the step change. The PRD shall stay in that condition for the entire exposure period.

After removing the source, the PRD shall return to the state prior to the test within 1 min.

6.7 Detection of neutrons (if provided)

6.7.1 Requirements

The PRD shall trigger a neutron alarm within 5 s when exposed to a moderated ^{252}Cf source (see 5.3) with an emission rate (N_R) of $2 \times 10^4 \text{ s}^{-1}$ ($\pm 20\%$) at a distance (D_R) of 25 cm from the instrument's reference point.

6.7.2 Method of test

Using the test method described in 6.2.2, move the moderated ^{252}Cf source specified in the requirement at a speed of $1,2 \text{ m}\cdot\text{s}^{-1}$ towards the PRD and stop at the distance of closest approach of 25 cm.

Neutron alarms shall activate not later than 5 s after the source stops at the point that is closest to the PRD. The response is also acceptable if the neutron alarm is activated before the maximum radiation field is reached (i.e., before reaching the required test position). The source shall be removed and the test repeated 9 additional times for a total of 10 trials. Wait 100 s between trials. The test result is acceptable if the neutron alarm activates in at least 8 of 10 exposures.

If a source with emission rate of $2 \times 10^4 \text{ s}^{-1}$ ($\pm 20\%$) is not available, a moderated ^{252}Cf source (see 5.3) with an emission rate N_T placed at a distance D_T ranging between 20 cm and 50 cm may be used. In order to obtain the same neutron fluence rate at the point of measurement ($2,55 \text{ cm}^{-2}\cdot\text{s}^{-1}$), the testing distance shall be calculated as:

$$D_T = D_R \sqrt{N_T / N_R}$$

Gamma alarms are allowed during this test, but they will not be considered.

NOTE 1 The emission rate of $2 \times 10^4 \text{ s}^{-1}$ at 25 cm is approximately equivalent to a neutron fluence rate at the point of measurement of $2,55 \text{ cm}^{-2}\cdot\text{s}^{-1}$.

NOTE 2 The distance range from 20 cm to 50 cm corresponds to a N_T of between $1,28 \times 10^4 \text{ s}^{-1}$ and $8 \times 10^4 \text{ s}^{-1}$.

6.8 Gamma response of neutron detector (if provided)

6.8.1 Requirements

The PRD shall not trigger a neutron alarm when exposed to a ^{137}Cs gamma ray source producing an ambient dose equivalent rate at the reference point of the instrument of $100\ \mu\text{Sv}\cdot\text{h}^{-1}$ ($\pm 20\%$) for duration of 100 s. The neutron response shall not be affected by the presence of gamma radiation.

6.8.2 Method of test

Expose the PRD for 100 s to a ^{137}Cs radiation source producing an ambient dose equivalent rate of $100\ \mu\text{Sv}\cdot\text{h}^{-1}$ at the reference point of the instrument. The distance between the source and the instrument shall be not less than 25 cm. No neutron alarms should be observed during the test.

Repeat the neutron alarm test in 6.7 while the PRD is being exposed to the gamma radiation field. The test result is acceptable if the neutron alarm activates in at least 8 trials out of 10.

If the PRD over-ranges at the above dose rate value and the neutron alarm cannot be verified, then the ambient dose equivalent rate should be reduced to the maximum value for which the over-range condition is no longer present and the test should be carried out as described above.

7 Environmental requirements

7.1 Ambient temperature

7.1.1 Requirements

Over the temperature range specified in IEC 62706:2012 for body worn devices (from $-10\ ^\circ\text{C}$ to $+40\ ^\circ\text{C}$), the PRD shall function correctly and alarm at an increase of the ambient dose equivalent rate of $0,5\ \mu\text{Sv}\cdot\text{h}^{-1}$. As a result of the temperature change, the variation of the indication of the magnitude of the radiation field or of the ambient dose equivalent rate (if provided) shall be within the values described in the functionality test.

If the manufacturer has stated a wider operational temperature range, the instrument's ability to perform correctly shall be tested over that range.

7.1.2 Method of test

Perform the functionality pre-test as described in 5.6.2.

Follow the ambient temperature method of test specified in IEC 62706:2012 for both ramp (7.2.2) and shock change (7.3.2).

Following the temperature equilibrium time, at the low and high temperature extremes perform the functionality post-test as described in 5.6.3.

7.2 Relative humidity

7.2.1 Requirements

Over the range of relative humidity specified in IEC 62706:2012 for body worn devices (40 % to 93 % at $35\ ^\circ\text{C}$, currently), the PRD shall function correctly and alarm at an increase of the ambient dose equivalent rate of $0,5\ \mu\text{Sv}\cdot\text{h}^{-1}$. As a result of the humidity change, the indication of the magnitude of the radiation field or of the ambient dose equivalent rate (if provided) shall not change as described in the functionality test.

7.2.2 Method of test

Perform the functionality pre-test as described in 5.6.2.

Follow the relative humidity method of test specified in IEC 62706:2012, 7.4.2.

Following the temperature and relative humidity equilibrium time, at the low and high relative humidity extremes perform the functionality post-test as described in 5.6.3.

7.3 Moisture and dust protection

7.3.1 Requirements

The PRD case design shall meet the IP classification requirements stated in IEC 62706:2012 for body worn devices (IP 54, currently).

After the dust and moisture tests, the instrument shall function correctly and alarm at an increase of the ambient dose equivalent rate of $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$. The indication of the magnitude of the radiation field or of the ambient dose equivalent rate (if provided) shall not change as described in the functionality test.

Instruments that are specified to be resistant to salt water shall exhibit this same performance after being subjected to a salt water spray.

7.3.2 Method of test

Perform the functionality pre-test as described in 5.6.2.

The method of test shall follow the IP requirements stated in IEC 62706:2012, 7.6.2.

Following exposure, an inspection shall be performed to determine the extent of dust ingress. Particular attention shall be paid to the battery compartment and any other easily accessed portions of the instrument. The protection is satisfactory if, on inspection, powder has not accumulated in a quantity or location such that, as with any other kind of dust, it could interfere with the correct operation of the instrument or impair safety.

For instruments that are designed for use in salt water spray environments, the PRD shall exhibit the same performance after being subjected to a salt water spray as defined in IEC 60068-2-11 using a conditioning duration of 16 h. The PRD does not have to be powered during the test.

Perform the functionality post-test after the above disturbances as described in 5.6.3.

8 Mechanical requirements

8.1 Drop test

8.1.1 Requirements

The PRD shall withstand drops as specified in the drop test part of IEC 62706:2012, 8.2 for body worn instruments (drops from 1 m onto a concrete surface on each of their six sides). After the drops, the PRD shall function correctly and alarm at an increase of the ambient dose equivalent rate of $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$.

8.1.2 Method of test

Perform the functionality pre-test as described in 5.6.2.

Follow the drop requirement and method of test specified in IEC 62706:2012, 8.2 for body worn instruments.

Perform the functionality post-test after the drops as described in 5.6.3. The PRD is allowed to alarm during the drop.

8.2 Vibration test

8.2.1 Requirements

The PRD shall withstand vibration as specified in IEC 62706:2012, 8.3 for body worn instruments (random vibration with spectral density of $0,01 \text{ g}^2 \cdot \text{Hz}^{-1}$ with endpoints of 5 Hz and 500 Hz). After vibrations, the instrument shall function correctly and alarm at an increase of the ambient dose equivalent rate of $0,5 \mu\text{Sv} \cdot \text{h}^{-1}$.

8.2.2 Method of test

Perform the functionality pre-test as described in 5.6.2.

Follow the vibration requirement and method of test specified in IEC 62706:2012, 8.3 for body worn instruments.

Perform the functionality post-test after the vibration as described in 5.6.3.

9 Electric and electromagnetic requirements

9.1 Battery lifetime

9.1.1 Requirements

Each instrument should operate on rechargeable and/or non-rechargeable batteries. With new batteries of the type recommended by the manufacturer, the battery life time shall be greater than 100 h under no alarm conditions for instruments with non-rechargeable batteries, and greater than 16 h for instruments with rechargeable batteries. Under continuous alarm conditions, the battery lifetime should be greater than 30 min.

A low battery indicator shall be provided to inform the user that the batteries need to be replaced or recharged.

9.1.2 Method of test

Perform the functionality post-test as described in 5.6.3 after 100 h under no alarm conditions (16 h for units with rechargeable batteries) or after the manufactured stated maximal battery life time, whichever is greater. With new or recharged batteries activate all alarms (i.e., visual, audible, vibration) and observe the PRD for 30 min. The PRD should alarm continuously for at least 30 min.

Verify whether a low battery indicator is provided.

9.2 Electrostatic discharge, radiofrequency immunity, radiated emissions and magnetic fields

9.2.1 Requirements

9.2.1.1 Electrostatic discharge

The PRD shall function properly during and after exposure to ESD at 6 kV (contact discharge for conductive surfaces) or 8 kV (air discharge for non-conductive or insulating surfaces).

9.2.1.2 Radiofrequency immunity

The PRD shall function properly when exposed to RF fields in the ranges of 80 MHz to 1 GHz at $50 \text{ V}\cdot\text{m}^{-1}$ and 1,4 GHz to 2,5 GHz at $10 \text{ V}\cdot\text{m}^{-1}$ using 80 % amplitude modulation with a 1 kHz sine wave.

9.2.1.3 Radiated emissions

The electromagnetic fields emitted by the PRD at 3 m shall be less than what is shown in IEC 62706:2012, 9.4.1 concerning the radiated emissions up to 2,5 GHz.

9.2.1.4 Magnetic fields

The PRD shall be fully functional when exposed to a $100 \text{ A}\cdot\text{m}^{-1}$ 50 Hz or 60 Hz magnetic field.

9.2.2 Method of test

For each of the requirements listed in 9.2.1, perform the functionality pre-test as described in 5.6.2.

Follow each of the requirements and methods of tests specified in IEC 62706:2012, Clause 9 for electrostatic discharge, radiofrequency immunity, radiated emissions and magnetic fields concerning body worn instruments.

Perform the functionality post-test as described in 5.6.3 after each test as follows: the indication check during the above EMC disturbances and the alarm check after all of them.

10 Documentation

10.1 Type test report

The manufacturer shall provide a report covering the type tests performed in accordance with the requirements of this document.

10.2 Certificate

The manufacturer shall provide a certificate or other documentation containing at least the following information:

- contacts for the manufacturer including name, address, phone number and email address;
- type of instrument, detector and types of radiation the instrument is designed to measure;
- if pressurized detectors are used (e.g. ^3He), the manufacturer shall state the dimensions, the pressure and the active volume of the detector;
- range of gamma count rate and/or dose rate the instrument is designed to indicate;
- if applicable, range of neutron count rate and/or dose rate the instrument is designed to indicate;
- reference point and reference orientation for radiation sources used for calibration;
- location and dimensions of the sensitive volumes of the detectors;
- response of the instrument to different appropriate radiation energies;
- mass and dimensions of the instrument;
- power supply (battery) requirements, including battery lifetime;
- inclusion of any hazardous material that may be subject to regulation (such as radionuclide check source, pressurized gases, corrosive materials);
- declaration of conformity with respect to this document, IEC 62401.

10.3 Operation and maintenance manuals

The manufacturer shall supply an operational and maintenance manual containing at least the following information for the user:

- Operating instructions and restrictions.
- Orientation for use.
- Trouble-shooting guide.

Annex A (informative)

Summary of performance requirements

The summary of the most important performance requirements is shown in Table A.1.

Table A.1 – Summary of the most important performance requirements

Parameter	Performance requirement or specification	Sub-clause
Maximum size	15 cm x10 cm x 5 cm	4.2.1
Maximum mass	400 g	4.2.2
Acoustical alarm	A-weighted sound level between 85 dB(A) and 100 dB(A) at 30 cm from the alarm source.	4.2.3
Rate of false alarms	Up to 1 alarm in 1 h	6.1
Gamma alarm	Alarm within 2 s after exposure to an increase in the ambient dose equivalent rate level of $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$ (–0 %, +20 %) above background for ^{241}Am , ^{137}Cs and ^{60}Co when moving at a speed of $1,2 \text{ m}\cdot\text{s}^{-1}$ (± 10 %) at a distance of closest approach of 1,5 m.	6.2
Personal protection alarm	Alarm within 2 s of exposure to an increase in the ambient radiation level greater than personal protection alarm threshold.	6.3
Relative intrinsic error	No more than ± 50 % at $1 \mu\text{Sv}\cdot\text{h}^{-1}$, $10 \mu\text{Sv}\cdot\text{h}^{-1}$ and 50 % of the maximum stated range of indication for ^{241}Am , ^{137}Cs and ^{60}Co .	6.4
Detection of gradually increasing radiation	The alarm threshold shall not be affected by slowly increasing radiation ($0,1 \text{ m}\cdot\text{s}^{-1}$).	6.5
Over-range	Indication of an over-range condition when the ambient dose equivalent rate is greater than 2 times of the manufacturer-stated over-range condition.	6.6
Detection of neutrons (if provided)	Neutron alarm within 5 s when exposed to a moderated ^{252}Cf source with an emission rate of $2 \times 10^4 \text{ s}^{-1}$ (± 20 %) at a distance of 25 cm from the instrument's reference point.	6.7
Gamma response of neutron detector (if provided)	Shall not trigger a neutron alarm when exposed to a ^{137}Cs gamma ray source producing an ambient dose equivalent rate at the reference point of the instrument of up to $100 \mu\text{Sv}\cdot\text{h}^{-1}$ (± 20 %) for duration of 100 s.	6.8
COV for environmental, electromagnetic and mechanical tests	COV of each nominal mean reading ≤ 12 %.	5.2
Functionality test for environmental, electromagnetic and mechanical tests	a) Alarm at an increase of $0,5 \mu\text{Sv}\cdot\text{h}^{-1}$ over the ambient background level of radiation at least 19 times out 20 trials. b) The mean value shall not change by more than ± 30 % (for the unit indications) or one neighbouring indication change (for the unit-less indication).	5.6
Ambient temperature	Functionality test from -10 °C to $+40$ °C for ramp and shock change.	7.1
Relative humidity range	Functionality test from 40 % to 93 % at 35 °C.	7.2
Moisture and dust	IP 54	7.3
Mechanical drop	Withstand drops from 1 m onto a concrete surface on each of their six sides.	8.1
Vibration	Withstand random vibration with spectral density of $0,01 \text{ g}^2\cdot\text{Hz}^{-1}$ with endpoints of 5 Hz and 500 Hz.	8.2
Battery lifetime	At least 100 h for non-rechargeable and 16 h for rechargeable batteries	9.1
Electrostatic discharge	6 kV (contact discharge) or 8 kV (air discharge)	9.2.1.1
RF immunity	Function properly when exposed to RF fields in the ranges of 80 MHz to 1 GHz at $50 \text{ V}\cdot\text{m}^{-1}$ and 1,4 GHz to 2,5 GHz at $10 \text{ V}\cdot\text{m}^{-1}$ using 80 % amplitude modulation with a 1 kHz sine wave.	9.2.1.2
Radiated emissions	Emitted electromagnetic fields at 3 m shall be less than what is shown in IEC 62706 up to 2,5 GHz.	9.2.1.3
Magnetic fields	PRD shall be fully functional when exposed to a $100 \text{ A}\cdot\text{m}^{-1}$ 50 Hz or 60 Hz magnetic field.	9.2.1.4

Bibliography

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IEC 60846-1, *Radiation protection instrumentation – Ambient and/or directional dose equivalent (rate) meters and/or monitors for beta, X and gamma radiation – Part 1: Portable workplace and environmental meters and monitors*

IEC 61526, *Radiation protection instrumentation – Measurement of personal dose equivalents $H_p(10)$ and $H_p(0,07)$ for X, gamma, neutron and beta radiations – Direct reading personal dose equivalent meters*

IEC 62327, *Radiation protection instrumentation – Hand-held instruments for the detection and identification of radionuclides and for the estimation of ambient dose equivalent rate from photon radiation*

ISO 8529-3:1998, *Reference neutron radiations – Part 3: Calibration of area and personal dosimeters and determination of response as a function of energy and angle of incidence*

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